

Running times of loops

Write a small Python script to measure the running time of loops. First, define a number n . Before each loop define a counter variable and set it to zero. In each step of the loop increment the counter by 1. After the loop has finished, print the counter. Implement the following loops:

- 1) **Loop A.** Write a **while** loop that starts with the loop variable being equal to n and runs until the loop variable is larger than 0. Within the loop integer divide the loop variable by two.
- 2) **Loop B.** Write a single **for** loop from 0 to $n - 1$.
- 3) **Loop C.** Write a nested loop with Loop B as the outer loop and Loop A as the inner loop.
- 4) **Loop D.** Write a nested loop with Loop B as the outer loop and Loop B as the inner loop, i.e., a nested loop with two **for** loops.
- 5) **Loop E.** Write a nested loop with Loop D as the outer loop and Loop A as the inner loop, i.e., Loop A is enclosed in two **for** loops.
- 6) **Loop F.** Write a nested loop with three **for** loops.

Instructions: In your code, write comments after printing the results and discuss the running times you have obtained.

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